

**Cool summers are stormy summers: precession reorganizes extratropical
cyclone activity through the quiet season**

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7 ABSTRACT: Extratropical cyclones peak in winter, while orbital precession reshapes the sea-
8 sonal insolation cycle most strongly in summer. Thus storm tracks might be expected to ignore the
9 precession cycle. This is examined with twelve equilibrium simulations using a coupled climate
10 model AWI-ESM2, spanning a full precession cycle at 30° longitude increments under fixed eccen-
11 tricity and obliquity. Extratropical cyclones are explicitly tracked based on 6-hourly atmospheric
12 pressure fields for both hemispheres. While the annual-mean cyclone response to precession is
13 muted, substantial seasonal variability emerges. In the North Atlantic, the North Pacific, and the
14 Southern Ocean the response concentrates in local summer, where storm activity ranges across
15 the phases by up to 68% of its climatological mean, while winter changes stay below about 10%.
16 We also find summers spent near perihelion feature suppressed storminess, whereas summers near
17 aphelion experience intensified cyclone activity, yielding antiphase midlatitude storm variability
18 between the two hemispheres over the precession cycle. This signal is carried by cyclone number
19 and position, and it follows the pronounced summer response of the Eady growth rate driven
20 primarily by vertical wind shear in all three basins. Precession therefore shapes extratropical storm
21 tracks mostly through their quiet season (i.e., summer). Summer-sensitive proxy archives should
22 record a strong and hemispherically antiphased storminess signal at the precession period that
23 annually integrating archives would miss. Moreover, monthly storm responses are phase-locked
24 to the calendar timing of perihelion, lagged by 1-2 months due to surface thermal inertia. This
25 fingerprint also appears when we apply the same cyclone tracking algorithm to pre-industrial (PI, 0
26 ka, perihelion falling in winter), Last Interglacial (LIG, 127 ka, perihelion falling in summer) and
27 mid-Holocene (MH, 6 ka, perihelion falling in autumn), with weaker amplitudes matching their
28 weaker eccentricity.

29 SIGNIFICANCE STATEMENT: Long orbital cycles drive global climate variability, and the
30 strongest of these, precession, reshapes sunlight mainly in summer, while mid-latitude storms
31 happen mainly in winter. This creates an apparent disconnect between precessional forcing and
32 storm track variability. Climate model simulations covering a full precession cycle show that
33 the orbit nevertheless controls storminess by acting through summer. When perihelion falls in a
34 hemisphere's summer, that summer has fewer mid-latitude storms and the storm belt sits farther
35 poleward, and the two hemispheres evolve in antiphase through the precession cycle. Simulations
36 of the Last Interglacial, when perihelion falls in northern summer, reproduce this pattern. This
37 tells researchers where to look for orbital storm signals in geological records, namely in archives
38 sensitive to summer, not to the yearly average.

39 1. Introduction

40 Extratropical cyclones govern midlatitude weather systems. They transport the majority of pole-
41 ward heat and moisture outside the tropics, supply a substantial fraction of midlatitude precipitation,
42 and aggregate into the canonical storm-track pathways of the North Atlantic, North Pacific, and
43 Southern Ocean (Hoskins and Valdes 1990; Chang et al. 2002; Hoskins and Hodges 2002, 2005).
44 These cyclones develop via baroclinic instability driven by the meridional temperature gradient
45 (Eady 1949), a gradient that reaches its maximum steepness during winter. Cyclone activities
46 consequently peak in the cold season, with consistent wintertime maxima (e.g., frequency) evi-
47 dent in both the Northern Hemispheres (Hoskins and Hodges 2002) and the Southern Hemisphere
48 (Simmonds and Keay 2000). Midlatitude cyclogenesis is therefore most favorable during winter.

49 Precession, by contrast, exerts its primary forcing on warm seasons. Slow variations in the
50 longitude of perihelion represent the leading orbital control on seasonal insolation cycles. It
51 redistributes incoming solar radiation, and the amplitude of insolation anomalies scales with a
52 season's background solar irradiance (Berger 1978; Laskar et al. 2004). Midlatitude summertime
53 insolation fluctuates by over 100 W m^{-2} across a full precession cycle, whereas midwinter insolation
54 experiences only minor orbital modulation. This stark contrast explains why orbital climate
55 frameworks single out summertime solar forcing as the most impactful component of climate
56 changes (Huybers 2006). Since extratropical cyclone activity is inherently winter-favored, this

invites a simple expectation that precession should exert negligible control over midlatitude storm tracks. But this expectation has never been put to a clean test.

Previous studies have looked closely at precessional signals, and they find seasonal shifts in climate. This is well established in monsoon research, from the classic orbital experiments of Kutzbach and Guetter (1986) to single-forcing designs that isolate precession explicitly (Bosmans et al. 2015; Merlis et al. 2013), and cave records show the northern and southern monsoons swinging in antiphase as perihelion migrates through the year (Wang et al. 2008). The extratropics have received far less attention. Kaspar et al. (2007) simulated a stronger and poleward-shifted North Atlantic winter storm track for the Eemian from two orbital time slices. Brayshaw et al. (2010) linked Holocene orbital shifts to North Atlantic and Mediterranean winter storm tracks via changes in baroclinicity. And Varma et al. (2012) found the Southern Hemisphere westerlies migrating under Holocene orbital forcing. Explicit cyclone tracking has entered paleoclimate work mainly at the Last Glacial Maximum, where ice sheets dominate the forcing (Pinto and Ludwig 2020; Raible et al. 2021), and in transient simulations of the last millennium (Hess and Chemke 2025). For tropical cyclones the precessional response has recently been isolated in high-resolution paleoclimate simulations, where it is governed by atmospheric moisture (Raavi et al. 2023). Here our study provides the extratropical counterpart, where cyclone growth is baroclinic. To our knowledge, no previous study has applied explicit cyclone tracking across a full precession cycle to isolate the orbital response of the extratropical storm tracks, and none has asked which season carries the response.

This missing piece of research matters beyond just storm dynamics. Many paleoclimate archives that record past storm activity such as wind-blown dust and sediment, or rainfall linked to westerly winds only capture signals from specific seasons instead of annual averages. If a proxy record mainly reflects one single season, it can either hide or falsely create an orbital signal, depending on which time of year carries the strongest storm response (Laepple et al. 2011). Fully understanding which season holds the clearest precessional storm signal is therefore essential for properly interpreting these geological records. Moreover, different metrics for cyclone behavior, e.g., total storm counts, how much rain each storm produces, and storm intensity, react differently to orbital forcing. Researchers must evaluate these quantities separately before drawing conclusions from any single geological proxy.

87 To fill this gap, we run 12 equilibrium climate simulations covering a full precession cycle.
88 Every extratropical cyclone in both hemispheres is tracked individually using 6-hour sea level
89 pressure. Our experimental design isolates precession at amplified eccentricity with all other
90 boundary conditions fixed, so that the seasonal storm response to precession forcing alone can be
91 examined. The results are then tested in simulations of real orbital states, such as PI, MH, and
92 LIG. Section 2 describes our model and simulation, and the method we use for storm tracking and
93 analysis. Section 3 presents the results. We discuss and conclude in section 4 and 5 respectively.

94 **2. Methodology**

95 *a. Model and experiments*

96 The simulations are performed with the coupled Earth system model AWI-ESM2 (Sidorenko
97 et al. 2019). The atmosphere component is ECHAM6 at T63 resolution with 47 vertical levels
98 (Stevens et al. 2013), which includes the land surface module JSBACH (Raddatz et al. 2007). The
99 ocean and sea ice component is FESOM2, a finite-volume model formulated on an unstructured
100 mesh (Danilov et al. 2017), with a spatially varying resolution from 25 km over high latitudes and
101 coastal regions, to 150 km over open oceans.

102 12 equilibrium simulations are analyzed, taken from the idealized precession ensemble introduced
103 by Shi et al. (2025). The longitude of perihelion ω is prescribed at values from 0° to 330° in steps of
104 30° (labeled as p000, p030..., and p330), while eccentricity is fixed at 0.058 and obliquity at 24.5° .
105 The eccentricity value sits near the upper bound reached during the Quaternary and amplifies the
106 precessional modulation of insolation. All remaining boundary conditions follow the pre-industrial
107 setup, and each phase is branched from a long pre-industrial control. 30 years of 6-hourly output
108 from the equilibrated part of each simulation form the basis of the cyclone analysis. The calendar
109 month of perihelion for each phase is computed from orbital geometry (Berger 1978), and the same
110 formulary provides the insolation fields shown in Fig. 1.

111 To test our results under realistic orbital configurations, we additionally conducted three equilib-
112 rium simulations including a PI (0 ka) control run, a MH (6 ka) experiment, and a LIG (127 ka)
113 experiment. Boundary conditions follow the protocols of PMIP4 (Otto-Bliesner et al. 2017). Or-
114 bital parameters are calculated following Berger (1977), and the greenhouse gas concentrations are
115 taken from multi-archive reconstructions from the Greenland and the Antarctic ice cores (Flückiger

et al. 2002; Monnin et al. 2004; Schneider et al. 2013; Schilt et al. 2010; Buiron et al. 2011). Each simulation is integrated for 1,000 model years with the trend of simulated global mean surface temperature in the final 100 model years not exceeding ± 0.05 K/century. We preserved 6-hourly sea level pressure from each simulation. At 127 ka, perihelion occurs close to the June solstice with an eccentricity of 0.039378; at 6 ka, perihelion falls within September, while the pre-industrial control features perihelion in early January.

122 *b. Cyclone detection*

Extratropical cyclones are detected and tracked with the algorithm of Crawford et al. (2021), and Fig. 2 summarizes the procedure with one 6-hourly field of the ensemble. 6-hourly sea level pressure is first interpolated onto 100-km equal-area grids centered on each pole, and the two hemispheres are processed separately. A grid cell qualifies as a cyclone center when its pressure p_c lies below that of all 8 neighbors and the surrounding field keeps a sufficient inward gradient. The gradient test compares p_c with the mean pressure $\bar{p}(d)$ on a one-cell ring at distance d from the candidate,

$$\frac{\bar{p}(d) - p_c}{d} \geq \gamma \quad (1)$$

where $d = 1000$ km and $\gamma = 750 \text{ Pa} (1000 \text{ km})^{-1}$, and candidates over terrain above 1500 m are discarded. Centers closer than 1000 km are treated as one multicenter system. The cyclone area A is obtained by growing closed isobars around each center at a 200-Pa interval until a contour fails to close, and the last closed isobar defines the edge pressure p_e (Fig. 2b). Each cyclone carries the depth and the area-equivalent radius

$$D = p_e - p_c \quad (2)$$

$$R = \sqrt{A/\pi} \quad (3)$$

Tracking links the centers of consecutive 6-hourly fields (Fig. 2c). For every active cyclone a first-guess position is projected from its previous displacement, damped to three quarters,

$$\mathbf{x}_q = \mathbf{x}_t + 0.75 (\mathbf{x}_t - \mathbf{x}_{t-1}) \quad (4)$$

137 and a center of the next field qualifies as the continuation when it lies within $v_{\max}\Delta t$ of both the
 138 current position $\mathbf{x}_t$ and the projection $\mathbf{x}_q$, with $v_{\max} = 150 \text{ km h}^{-1}$ and $\Delta t = 6 \text{ h}$, a searching radius
 139 of 900 km per step. Among the qualifying candidates the one nearest to the projection continues
 140 the track. Centers left unmatched at the new time start tracks as genesis events, and tracks left
 141 unmatched end as lysis. Only systems poleward of 30° latitude enter the analysis.

142 Several track-based diagnostics are derived. Track density counts every 6-hourly cyclone position,
 143 system density counts each cyclone once at its mean position, and genesis density counts the first
 144 point of each track. Tracks that merely cross the boundaries of the monthly processing windows
 145 would appear as spurious genesis or lysis events, and such truncated events are removed. For every
 146 cyclone the lifespan, the maximum depth relative to the outermost closed isobar, the minimum
 147 central pressure, the mean cyclone area, and the propagation speed are recorded. We further
 148 define a storm-day field, representing the number of days per month during which a grid point lies
 149 inside a detected cyclone. It therefore measures the storm-influence time of each grid point, and
 150 it is affected by most cyclone metrics, such as the number of cyclones, their area, lifespan, and
 151 propagation speed, though not by their intensity. Frequency statements in this paper therefore rest
 152 on track density.

153 *c. Baroclinicity diagnostics*

154 The maximum Eady growth rate serves as the measure of baroclinicity (Eady 1949; Lindzen
 155 and Farrell 1980; Hoskins and Valdes 1990). It is computed from monthly wind, temperature, and
 156 geopotential height in the 925–700 hPa layer as

$$\sigma = 0.31 |f| \frac{|\partial U / \partial z|}{N}, \quad (5)$$

157 where f is the Coriolis parameter, $|\partial U / \partial z|$ the vertical wind shear, and N the Brunt–Väisälä
 158 frequency. The response to precession is quantified between the end members p090 and p270,
 159 which place perihelion at the June and December solstices. Because the difference fields are
 160 dipolar, plain box averages cancel, and the response amplitude is instead defined as the root-mean-
 161 square of the difference $\Delta\sigma = \sigma_{90} - \sigma_{270}$ over a basin, normalized by the climatological growth
 162 rate of the season.

163 To separate the roles of shear and stratification we follow the substitution approach of Camargo
 164 et al. (2007). Writing $\sigma = 0.31 |f| S/N$ with $S = |\partial U / \partial z|$, the difference is split by changing one
 165 factor at a time while holding the other at its p270 value,

$$\Delta\sigma_S = 0.31 |f| \frac{S_{90} - S_{270}}{N_{270}}, \quad \Delta\sigma_N = 0.31 |f| S_{270} \left(\frac{1}{N_{90}} - \frac{1}{N_{270}} \right), \quad (6)$$

166 so that $\Delta\sigma = \Delta\sigma_S + \Delta\sigma_N + \varepsilon$, with a small cross term ε . The share of each factor is the projection
 167 of its field onto the full difference pattern over the basin sector,

$$c_S = \frac{\langle \Delta\sigma_S \Delta\sigma \rangle}{\langle \Delta\sigma \Delta\sigma \rangle}, \quad c_N = \frac{\langle \Delta\sigma_N \Delta\sigma \rangle}{\langle \Delta\sigma \Delta\sigma \rangle}, \quad (7)$$

168 where $\langle \cdot \rangle$ denotes the area-weighted average over the storm-track box defined in the next sub-
 169 section.

170 *d. Basins and statistics*

171 Three storm-track basins are analyzed, the North Atlantic (30–75°N, 80°W–20°E), the North
 172 Pacific (30–75°N, 120°E–120°W), and the Southern Ocean (40–75°S, all longitudes). Basin means
 173 are area weighted. The size of the precessional response is reported as the range across the twelve
 174 phases, the maximum minus the minimum of the 30-year mean, expressed as a percentage of the
 175 phase mean. Uncertainty estimates use the interannual spread of the 30 individual years in each
 176 phase, and the significance of the end-member difference maps is assessed with a two-sided Welch
 177 t test (Welch 1947) at each grid point.

178 **3. Results**

179 *a. Summer sensitivity*

180 The 12 precession-phase annual midlatitude storm climatologies exhibit only minor differences,
 181 and phase anomalies relative to the multi-phase ensemble mean manifest as weak, slowly evolving
 182 dipoles aligned along major storm track pathways (Fig. 3). The basin-averaged time series quantify
 183 the muted amplitude of this annual-mean signal, with across-phase ranges of 10%, 17% and 4% in
 184 the North Atlantic, the North Pacific and the Southern Ocean, varying smoothly with ω . We also
 185 observe an interhemispheric antiphase.

186 There is also a pronounced seasonal cycle. Extratropical cyclone activities peak in DJF in North-
187 ern Hemisphere basins and JJA across the Southern Ocean, with this seasonal timing unchanged
188 across all orbital setups (Fig. 4a–c; Table 1). Cold-season cyclones also carry greater intensity,
189 with winter peak cyclone depths exceeding summertime values by 70-100 % within the Northern
190 Hemisphere storm basins (not shown). Overall, winter dominates midlatitude cyclone activity
191 uniformly across hemispheres and all orbital phases.

192 Precession exerts negligible control over annual-mean storm conditions but profoundly reshapes
193 seasonal cyclone variability, and its strongest modulation targets the low-activity warm season.
194 Though each precession phase retains the same wintertime peak in basin-averaged storm track
195 density, cross-phase deviations emerge almost entirely within each basin’s local summer (Fig. 4a–c).
196 Seasonal range statistics quantify this stark seasonal contrast in orbital sensitivity (Fig. 4d–f). For
197 the North Pacific, the across-phase range of track density accounts for just 7 % of the phase mean on
198 an annual basis, yet rises to 44 % in boreal summer (JJA). The North Atlantic follows an identical
199 pattern, with annual variability of 7 % versus a summertime range of 25 %. The Southern Ocean
200 mirrors this behavior but shifts the sensitive season to its local summer (DJF), with an annual range
201 of only 4 % compared to a 22 % summer signal. All three basins identify local summer as the
202 most orbitally sensitive season. The muted annual signal is not a diluted version of this summer
203 response but the residual of a partial cancellation, because summer and the colder seasons respond
204 with opposite sign. In the North Pacific the extreme phases differ by –44 % in summer yet by +3 %
205 in winter, and over the Southern Ocean the summer gain of +18 % is opposed by a winter loss of
206 –8 %, so the two largely offset once the year is averaged.

207 This summertime storm response is locked to orbital geometry. When basin-averaged track-
208 density anomalies are plotted against both calendar month and precession angle ω , the band of
209 peak anomalies traces a diagonal ridge aligned with the calendar timing of perihelion across
210 the full phase space (Fig. 4g–i), with a delay of 1-2 months. Summers spent near perihelion
211 exhibit suppressed storm activity, whereas aphelion summers feature intensified storminess. A
212 single precessional forcing thus drives opposing midlatitude storm responses between the two
213 hemispheres across the orbital cycle, as northern summer perihelion aligns with southern summer
214 aphelion under orbital precession, and vice versa, reversing the insolation control on warm-season
215 storminess between hemispheres.

216 *b. Mechanisms*

217 The pronounced seasonal dependence of orbital storm signals originates directly from insolation
218 forcing. Averaged across the Northern Hemisphere storm-track band (40–70°N), precessional
219 insolation variability peaks at 105 W m⁻² during boreal summer (JJA) and falls to merely 14 W m⁻²
220 in boreal winter (DJF). The Southern Hemisphere storm band (40–70°S) exhibits mirror-symmetric
221 behavior, with a maximum seasonal forcing amplitude of 119 W m⁻² in austral winter (DJF) versus
222 just 47 W m⁻² in austral summer (JJA) (Fig. 1). The underlying mechanism lies in the multiplicative
223 nature of precessional insolation modulation. Shifting perihelion across the annual cycle alters
224 the Sun–Earth distance for each month, which rescales monthly incoming solar flux by a factor
225 proportional to eccentricity e (approximately $4e$, equaling 23 % for $e = 0.058$). Midwinter at
226 midlatitudes features minimal climatological insolation, leaving little baseline flux to be modulated
227 by precessional shifts. Precession forcing therefore exerts its strongest modulation on local summer
228 in each hemisphere, and barely perturbs deep winter conditions.

229 Orbital insolation anomalies are translated into extratropical storm activity via shifts in midlat-
230 itude baroclinicity. Differences in maximum Eady growth rate between the extreme precession
231 phases $\omega = 90^\circ$ and $\omega = 270^\circ$ reach their peak magnitude within each basin’s local summer (Fig. 5).
232 Normalized against seasonal climatological means, summertime Eady growth rate responses hit
233 31 % in the North Atlantic, a pronounced 92 % in the North Pacific, and 21 % in the Southern Ocean.
234 A decomposition analysis further reveals that vertical wind shear dominates the precessional Eady
235 growth rate anomalies, while static stability contributes negligible signal strength. This is consis-
236 tent across all storm tracks. Perihelion summers warm the summer hemisphere, with the strongest
237 850-hPa warming over the subpolar continents and a weaker response over the ice-covered Arctic
238 (Fig. A1). The meridional temperature gradient therefore flattens along the equatorward flank of the
239 storm track and steepens along its poleward flank, the thermal counterpart of the growth-rate dipole.
240 The weakened shear suppresses the baroclinic growth that sustains warm-season cyclones, and the
241 steepened poleward flank supports the poleward shift. The storm anomalies appear somewhat
242 poleward of the growth-rate anomalies, consistent with the downstream development of baroclinic
243 waves (Chang et al. 2002).

244 *c. Meridional shift*

245 Spatial difference maps between the two orbital end-members (p090 minus p270) yields weak
246 responses for the annual means (Fig. 6a). By contrast, boreal summer (JJA) exhibits a prominent
247 dipole pattern across Northern Hemisphere basins, with fewer storm days through the midlatitude
248 core and more along the Arctic flank at perihelion summer, the signature of a poleward shift joined
249 to an overall summer weakening. In DJF the signal migrates to the Southern Ocean and forms a
250 circumpolar banded structure of opposite sign, marking enhanced midlatitude storm activity and
251 an equatorward shift of the southern storm belt.

252 The latitudinal displacement of storm tracks driving these dipole anomalies is quantified in Fig. 7.
253 For the North Pacific boreal summer, core storm latitude shifts from 53°N when perihelion falls in
254 northern winter to 61°N when it falls in northern summer, yielding an 8° total poleward migration
255 that peaks precisely at $\omega = 90^{\circ}$. The North Atlantic summertime storm track undergoes a 4.4°
256 poleward shift following identical orbital behavior, while the Southern Ocean mirrors this response
257 with a 3.8° poleward displacement when perihelion coincides with austral summer (near p270).
258 A perihelion summer therefore weakens storm activity and shifts the storm population poleward,
259 consistent with the poleward shift of baroclinic activity evident in the Eady growth rate dipole
260 (Fig. 5a).

261 By contrast, the annual-mean track latitude varies by no more than 1.3° in any basin. Winter
262 storm-track latitudes only fluctuate by $1.5\text{--}2.8^{\circ}$ across the full precession cycle, and their latitudinal
263 extrema do not align with solstitial orbital phases. This further confirms our earlier conclusion
264 that midlatitude winter storms are orbitally insensitive. Spring and autumn display intermediate
265 latitudinal variability with a coherent phase dependence. For example, autumn poleward displace-
266 ment maximizes near $\omega = 150^{\circ}$, the orbital configuration that places perihelion in late summer.
267 Latitudinal poleward shifts therefore track the calendar timing of perihelion identically to Fig. 4g–i,
268 with a uniform lag of 1–2 months, consistent with surface thermal inertia. Figure 8 quantifies this
269 basin-dependent lag. All basin-scale storm extrema cluster along basin-specific dashed lines offset
270 from the perihelion reference. The response lags the perihelion calendar by distinct intervals across
271 regions, near 1 month in the North Pacific, 1–2 months in the North Atlantic, and slightly over 2
272 months in the Southern Ocean.

273 *d. Cyclone precipitation*

274 The hydrological signature of precessional forcing evolves in antiphase with storm dynamical
275 responses (Fig. 9). Detected extratropical cyclones contribute roughly one-third of total midlatitude
276 precipitation across our simulations, consistent with storm precipitation partitioning derived from
277 observational reanalysis datasets (Hawcroft et al. 2012). During boreal summer over the North
278 Pacific, mean precipitation per individual storm rises by as much as 15 % at the orbital phase where
279 total storm frequency reaches its minimum, yielding nearly perfectly opposing variability between
280 the two metrics. Perihelion summers feature warmer, more humid midlatitude atmospheres, which
281 amplifies precipitation intensity within the remaining cyclones; nevertheless, the sharp reduction
282 in storm number dominates the total signal, such that basin-wide summertime precipitation drops
283 by 17 % between the two extreme orbital phases. The fractional share of precipitation originating
284 from cyclones also declines markedly, with a full-cycle range of 9.5 % in North Pacific summer
285 compared to just 3 % for the annual mean. Precessional forcing thus exerts two counteracting
286 controls on extratropical hydroclimate simultaneously. It reduces total cyclone counts while
287 boosting precipitation intensity per storm.

288 *e. Realistic paleosimulations (MH and LIG)*

289 Our results based on the 12 idealized simulations can be validated against the realistic orbital-
290 state simulations (PI, MH, and LIG) introduced in Section 2. In LIG, perihelion occurs near June
291 solstice, and the tracked cyclone responses fully reproduce our core theoretical pattern. Relative to
292 PI, summer storm track density decreases by 8% over the North Atlantic and 11% over the North
293 Pacific; winter cyclone changes remain within approximately 3%. Meanwhile, austral summer
294 storm track density across the Southern Ocean rises by 8%. Northern Hemisphere summer storm
295 tracks shift poleward by 1–2 degrees, while the Southern Hemisphere summer storm belt shifts
296 equatorward (Fig. 10). The summer difference composites show identical weakening and poleward
297 displacement of Northern Hemisphere storm tracks as seen in our amplified idealized end-member
298 simulations (Fig. 11). In MH, perihelion happens in September, leaving seasonal forcing dominated
299 by warm September conditions with cool March conditions. Consistent with this seasonal signal,
300 MH storm variability features fewer, poleward-shifted cyclones in autumn and the opposite trend
301 in spring. The magnitude of storm responses in these paleosimulations is much weaker than that

of our amplified end-member contrasts (p090 vs. p270, Fig. 6), consistent with their smaller eccentricity values of the real orbits. Notably, these experiments combine realistic precession with their own obliquity and greenhouse gases, so they test our theoretical prediction under multi-factor natural orbital forcing rather than isolating precessional effects.

4. Discussion

The seasonal preference seen in storm responses originates purely from orbital geometry, independent of model physics. Precession redistributes insolation in proportion to the insolation a season already has, meaning winter hemispheres have very little background insolation for orbital changes to modify (Berger 1978). Fig. 1 clearly illustrates this stark seasonal contrast in forcing strength. Winter insolation varies by only roughly 20 W m^{-2} across the precession cycle, while summer values exceed 125 W m^{-2} within mid-latitude storm-track latitudes. This is model independent and will hold in any climate models as it simply reflects the orbital geometry behind the well-known summer-dominated orbital forcing framework (Huybers 2006). Existing monsoon studies has long confirmed that summer bears the strongest precessional signal (Kutzbach and Guetter 1986; Bosmans et al. 2015). But the new insight of our work lies elsewhere. This summer-focused orbital forcing acts on a winter-dominated weather system, reshaping storm activity exclusively during its normally quiet warm season. Note that the term "quiet" here refers to the weaker overall intensity of the summer storm track, not its population. Both hemisphere still hosts abundant storm systems in local summer, and they are merely weaker, shallower and move more slowly compared to winter cyclones.

The mechanism that converts summer insolation into summer storms is the thermal wind. When perihelion falls in local summer, the corresponding hemisphere warms most strongly over the subpolar continents. The temperature gradient flattens on the equatorward side of the storm belt and steepens on its poleward side, so the vertical wind shear that drives baroclinic growth weakens where most storms grow and the growth zone moves poleward (Fig. A1). Our factor decomposition analysis confirms that the shear term carries essentially the whole Eady growth rate response in all three ocean basins, while static stability plays a negligible role. We see similar physical mechanism operating in the modern Northern Hemisphere, where greenhouse warming takes the place of perihelion-driven heating. Stronger summer warming at high latitudes has weakened summertime

331 large-scale circulation, eddy kinetic energy, and the number of strong summer cyclones (Coumou
332 et al. 2015; Chang et al. 2016; Gertler and O’Gorman 2019). Future climate projections show the
333 same shear-based decline of Northern Hemisphere summer cyclone activity and baroclinic energy
334 as the globe warms (Lehmann et al. 2014; Priestley and Catto 2022). But a difference should
335 be noted here, as greenhouse warming is a transient forcing that both hemispheres feel in phase,
336 whereas our simulations are equilibrated and precession swings both hemispheres in antiphase.
337 Even so, both forcings follow one unified rule that is the core theme of this study. Cooler summers
338 host more storm activity.

339 Compared to summer, storm systems in winter are much more stable. Across all three ocean
340 basins, winter storm activity changes by less than 10% over a full precession cycle, even though
341 winter Eady growth rates shift by 16–18% between the two orbital end-members, and weak winter
342 insolation forcing cannot fully explain this stability. The winter storm track appears saturated in the
343 sense that ample baroclinicity is available in every phase, so that moderate changes in the growth
344 rate do not translate into changes in occurrence. A transient simulation of the last millennium
345 likewise found cyclone counts nearly insensitive to external forcing (Raible et al. 2018; Hess and
346 Chemke 2025). Our findings expand this conclusion to the far larger swings in solar radiation
347 driven by precession, and this insensitivity only holds for wintertime conditions.

348 Geological records sensitive to summertime conditions within midlatitude storm belts should
349 preserve prominent precession-scale storm signals, whereas proxies that average climate signals
350 across winter or the full year will show only minor orbital imprint. Just as observed for temperature
351 reconstructions, the seasonal bias of a given archive can either generate artificial orbital storm
352 signals or fully obscure true precessional variations (Laepple et al. 2011). Furthermore, summer-
353 sensitive storm records from both hemispheres will fluctuate oppositely over precessional cycles,
354 forming an extratropical equivalent of the well-documented monsoon seesaw (Wang et al. 2008;
355 Erb et al. 2015). Current published sediment records already support the feasibility of testing
356 this framework. Distinct precession-period signals have been detected in Northern Hemisphere
357 midlatitude westerly proxies (Li et al. 2019), as well as North Pacific dust records. Precessional
358 variation appears specifically in moisture-sensitive dust fractions, while the annually averaged
359 dust flux responds primarily to obliquity forcing (Zhong et al. 2024). Meanwhile, dust intensity
360 archives that capture winter-only signals display very weak precessional power (Chen et al. 2014).

361 Regarding the Southern Hemisphere, subtropical margins of the southern westerly belt exhibit clear
362 precession cycles (Lamy et al. 2019), consistent with our results. And the presence or absence
363 of precession signals at individual sites depends on their latitudinal position within the wind belt
364 (Kaiser et al. 2024). Records of the southern westerlies also show orbital-band migrations of the
365 belt itself (Lamy et al. 2010; Varma et al. 2012), and glacial-state cyclone modeling has shown
366 how strongly reconstructed storminess depends on which cyclone property a proxy senses (Pinto
367 and Ludwig 2020).

368 Although the three storm basins follow the same physical mechanism, they exhibit different
369 regional behaviors, which can be distinguished through separate diagnostic metrics. In local
370 summer, storm frequency and track position vary substantially by 16–68% across the precession
371 cycle, while storm propagation speed remains largely unchanged. By contrast, storm intensity
372 exhibits clear basin-dependent differences. The Southern Ocean shows a 22% swing in storm
373 frequency but only a 4% change in storm intensity. The North Pacific features a systematic reduction
374 in overall storm activity during perihelion summers. This is consistent with previous findings
375 from anthropogenic warming scenarios that storm frequency and intensity respond to external
376 forcing through independent physical mechanism (Bengtsson et al. 2009; Ulbrich et al. 2009; Catto
377 et al. 2019). Hydrological responses further amplify this contrast. Individual storms produce
378 stronger precipitation in summers with reduced cyclone numbers, indicating that hydrological
379 adjustments exceed dynamical changes under orbital forcing, a pattern analogous to that identified
380 under greenhouse warming (Bengtsson et al. 2009). Given that extratropical precipitation is
381 predominantly generated by cyclones and frontal systems (Hawcroft et al. 2012; Catto and Pfahl
382 2013), this competing effect substantially alters precipitation partitioning. So a precipitation proxy
383 tend to record combined signal of reduced storm occurrence and intensified single-storm rainfall
384 during perihelion summers.

385 Our study is based on idealized simulations designed to isolate pure precessional forcing. The
386 experimental setup holds eccentricity, obliquity, ice sheets, and greenhouse gas concentrations
387 fixed and does not incorporate transient climate evolution or orbital cross-interactions. As such,
388 our results represent dynamical responses to idealized orbital conditions rather than reconstructions
389 of specific geological epochs. Future work could incorporate time-varying boundary conditions

390 and full orbital combinations to further generalize the storm seasonal response across broader
391 paleoclimate backgrounds.

392 **5. Conclusions**

393 In our study, we carry out a set of equilibrium climate simulations covering a full precession cycle,
394 with explicit tracking of all extratropical cyclones across both hemispheres to isolate precessional
395 forcing. We find that precession regulates extratropical cyclone primarily through local summer,
396 the season with naturally low storm activity, while winter storm tracks has only minor change
397 across all precession phases. Storm activity calms down when perihelion falls in local summer
398 and intensifies when aphelion coincides with local summer. And, as perihelion summer in one
399 hemisphere matches aphelion summer in the other, mid-latitude storminess of the two hemispheres
400 varies in antiphase during a precession cycle. This orbital signal propagates through shear-driven
401 baroclinicity and expresses itself in cyclones numbers and position, both of which are phase locked
402 to the calendar month of perihelion, with a delay of 1-2 months. Our result also bring clear
403 implications for paleoclimate proxy interpretation. Orbital storm signals should be extracted from
404 midlatitude archives sensitive to summer climate, and the recorded signals will display opposite
405 variations between the two hemispheres.

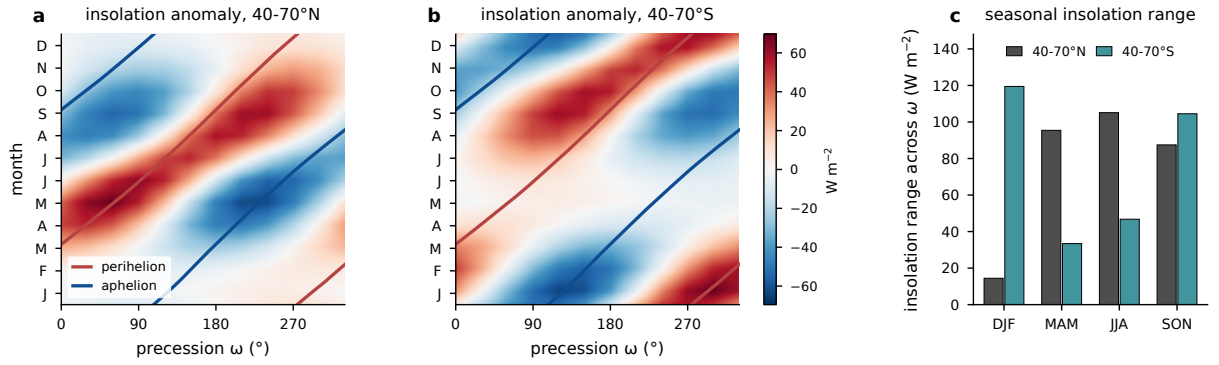


FIG. 1. Precessional insolation forcing from orbital geometry with eccentricity 0.058 and obliquity 24.5° .
 (a) Monthly insolation anomaly relative to the phase mean, averaged over the storm-track band $40\text{--}70^\circ\text{N}$, as
 a function of calendar month and precession phase ω , with the perihelion (red) and aphelion (blue) calendar
 months of each phase overlaid. (b) As in (a), but for $40\text{--}70^\circ\text{S}$. (c) Seasonal forcing amplitude in the two bands,
 the seasonal mean of the monthly maximum minus minimum insolation across the twelve phases. Units: W m^{-2} .

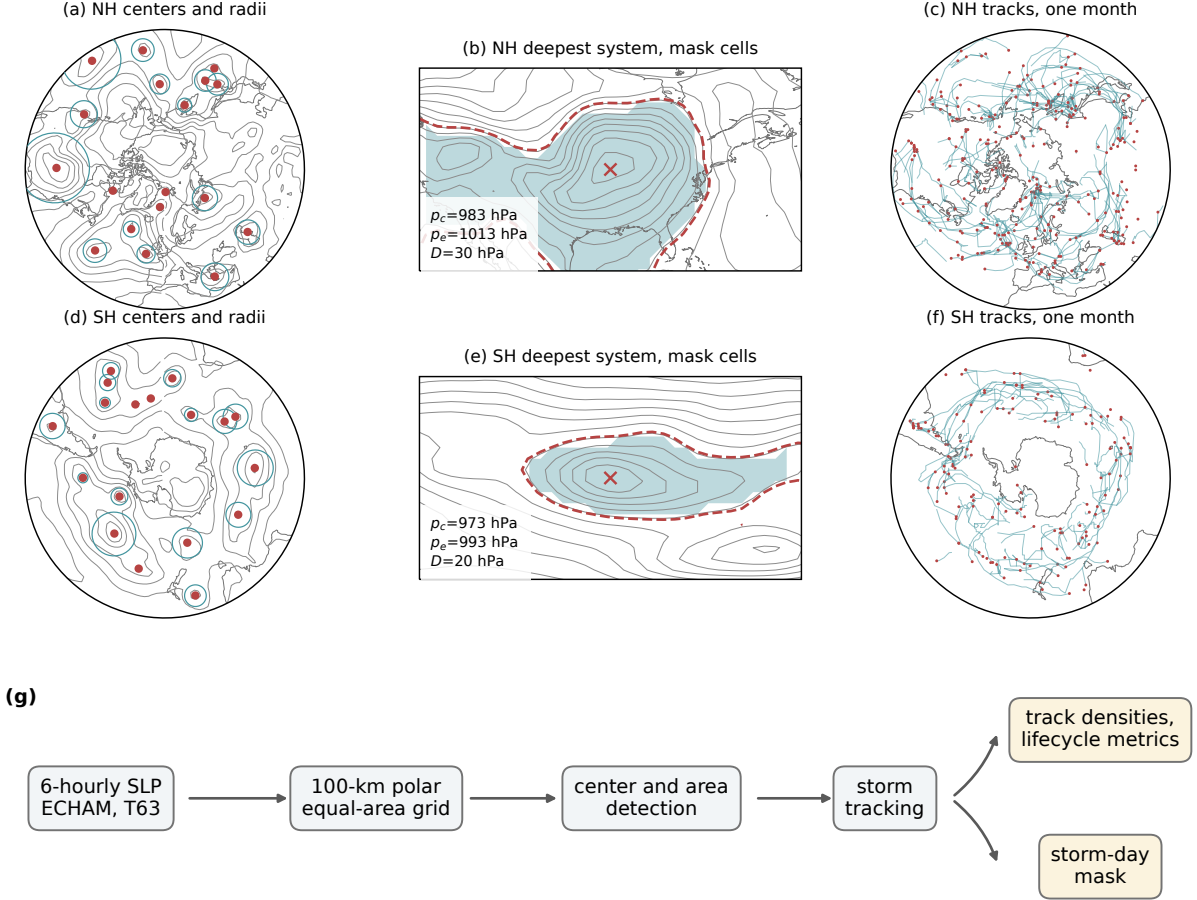


FIG. 2. Cyclone detection and tracking procedure. (a,c) One 6-hourly sea level pressure field of phase p000 (contours, 8-hPa interval) with all detected cyclone centers poleward of 30°N (dots) and their area-equivalent radii $R = \sqrt{A/\pi}$ (circles), 15 January of one model year. (b,e) The deepest system of that field, with the outermost closed isobar p_e (dashed), the cyclone center (cross), the central and edge pressures and depth annotated, and the binary mask cells (shading) defined by $\text{SLP} < p_e$ inside the $\sqrt{A}$ search window; contour interval 4 hPa. (c,f) All cyclone tracks of the same month (lines) with genesis points (dots). (d) Data processing workflow.

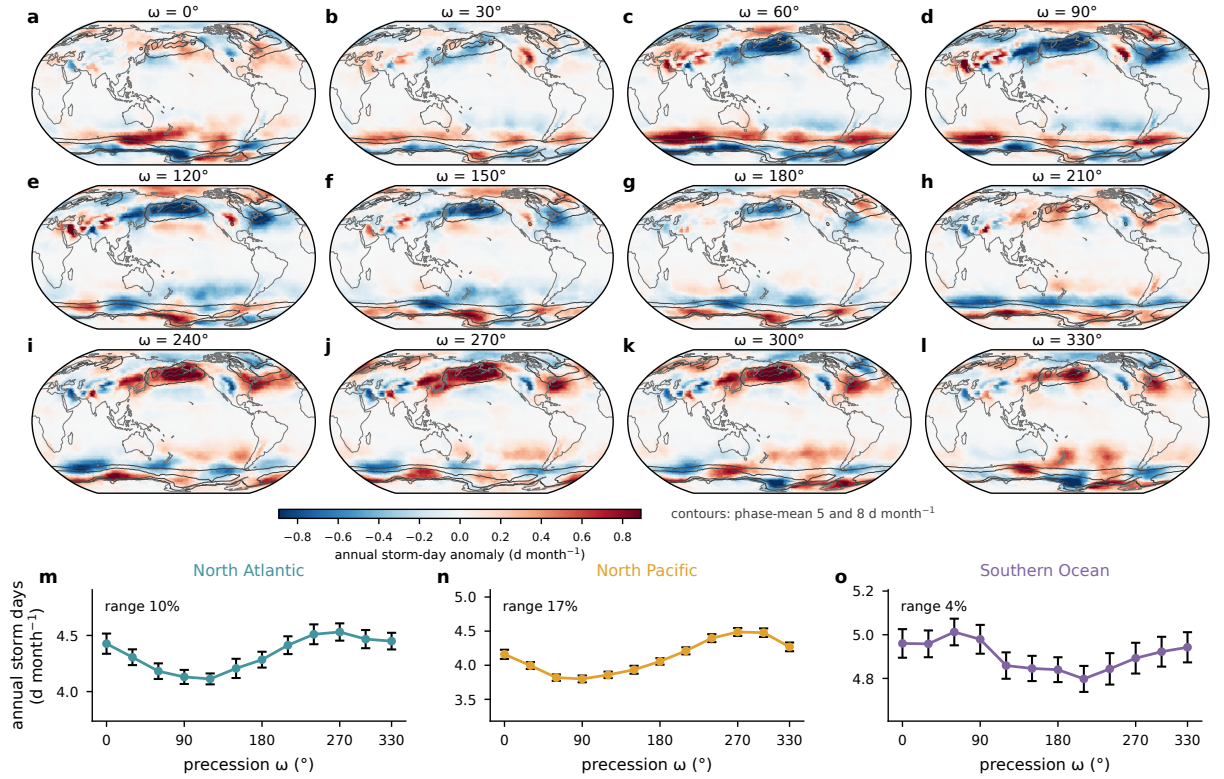


FIG. 3. Annual-mean storm days across the precession cycle. (a)–(l) Annual storm-day anomaly of each phase relative to the twelve-phase mean (shading), with the phase-mean climatology outlined at 5 and 8 d month⁻¹ (contours). (m)–(o) Basin-mean annual storm days against ω with interannual standard errors; the across-phase range as a percentage of the phase mean is annotated in each panel.

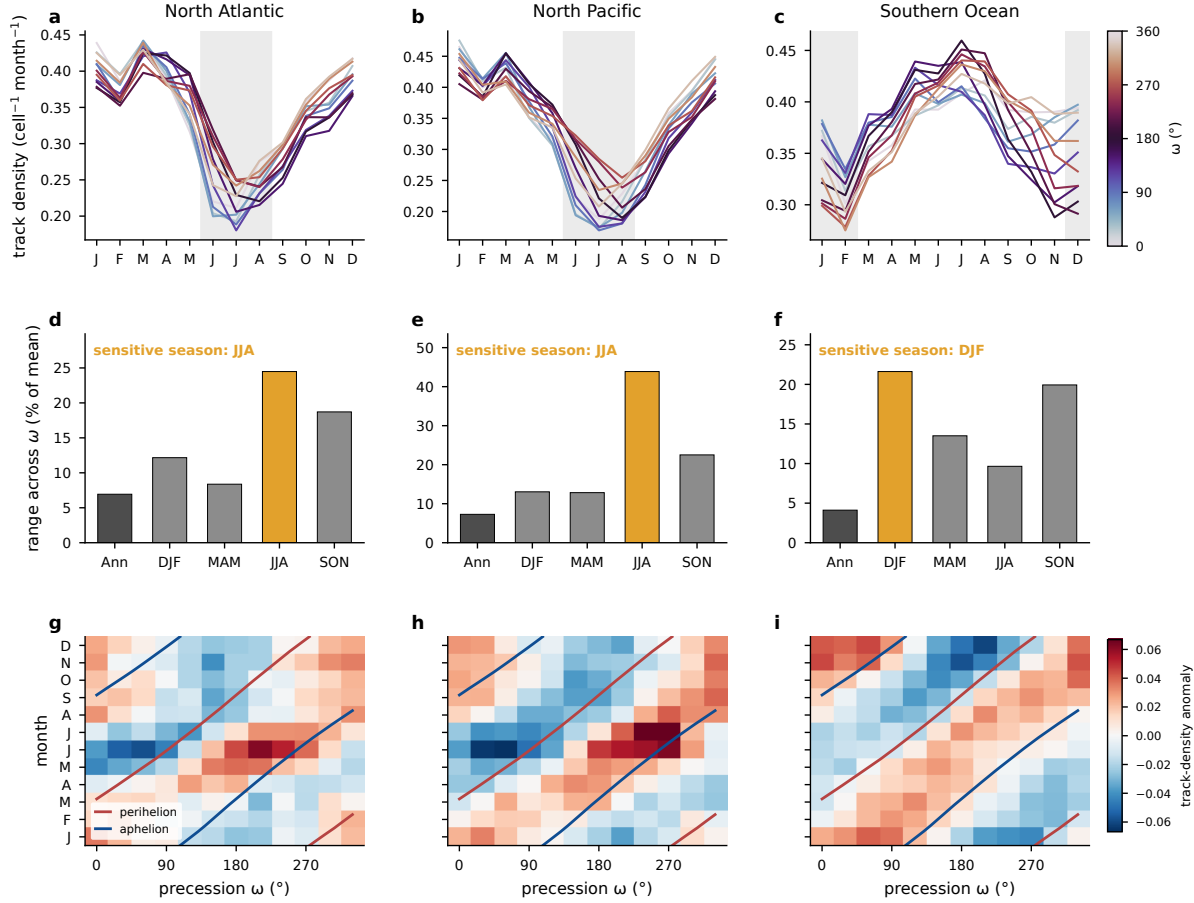


FIG. 4. Seasonal concentration of the precessional storm response, from the Lagrangian tracks. (a)–(c) Basin-mean track density (cell⁻¹ month⁻¹) by calendar month for the North Atlantic, the North Pacific, and the Southern Ocean, one line per phase colored by ω , with local summer shaded. (d)–(f) Across-phase range of track density (maximum minus minimum, percent of the phase mean) for the annual mean and the four seasons, with the local-summer bar highlighted; open symbols give the corresponding ranges of system and genesis densities. (g)–(i) Basin-mean track-density anomaly relative to the phase mean as a function of calendar month and precession phase (unsmoothed), with the perihelion (red) and aphelion (blue) calendar months of each phase overlaid.

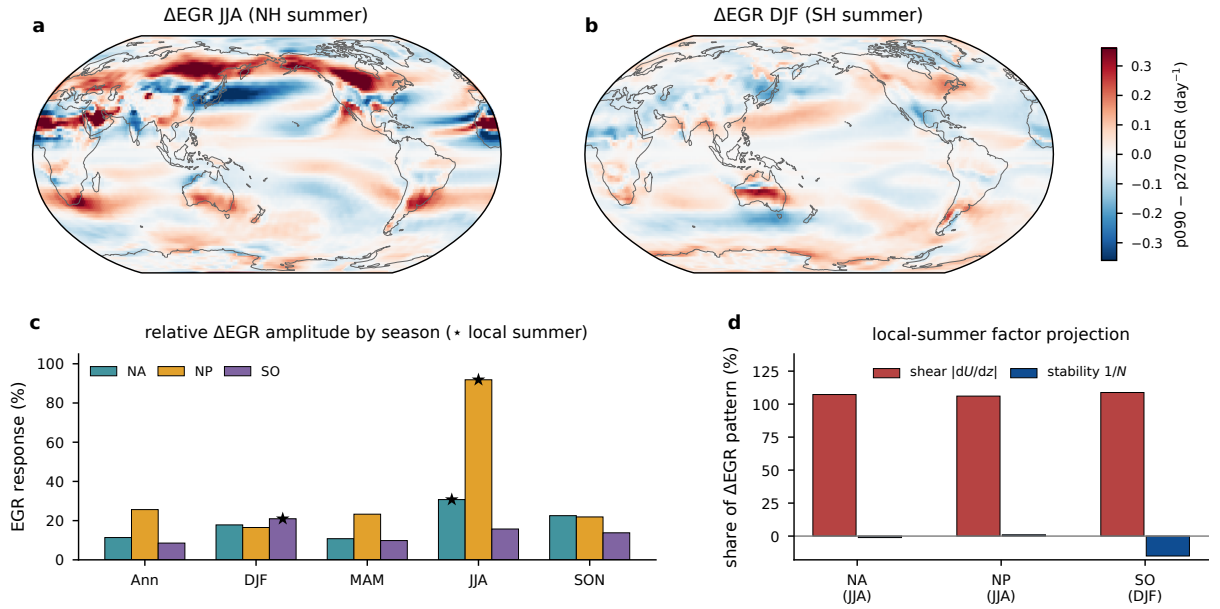


FIG. 5. Precessional response of the maximum Eady growth rate (925–700 hPa). (a) p090 minus p270 difference for JJA and (b) for DJF (day^{-1}). (c) Seasonal response amplitude in each basin, the RMS of the p090 minus p270 difference over the storm box as a percentage of the seasonal climatology, with stars marking local summer. (d) Projection shares of the vertical-shear and static-stability factors in the local-summer difference pattern.

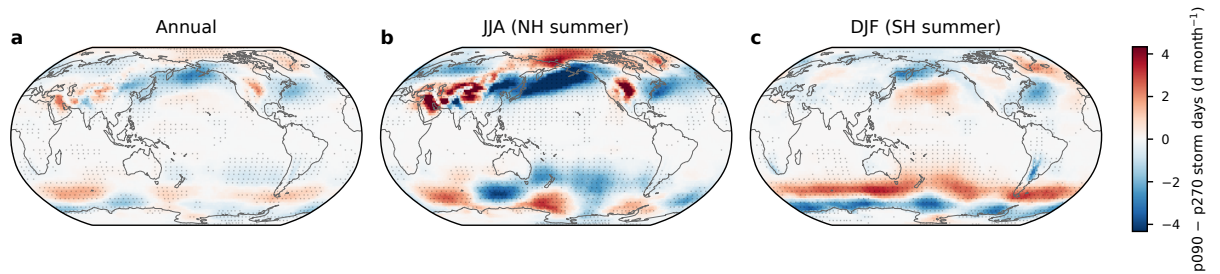


FIG. 6. Difference in storm days (d month^{-1}) between p090 and p270 for (a) the annual mean, (b) JJA, and (c) DJF, on a common color scale. Stippling marks grid points where the difference is significant at the 95% level based on a two-sided Welch t test over the 30 simulated years.

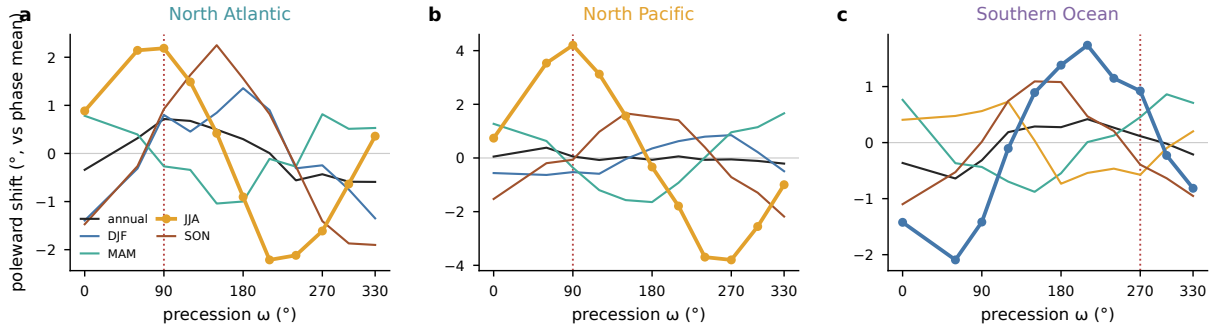


FIG. 7. Storm-track latitude across the precession cycle, the storm-day-weighted mean latitude over each basin sector shown as the poleward anomaly from each season's phase mean, for (a) the North Atlantic, (b) the North Pacific, and (c) the Southern Ocean. Lines give the annual mean and the four seasons, with the local summer of each basin drawn heavy with markers. Vertical dotted lines mark the phase with perihelion in the local summer of each basin.

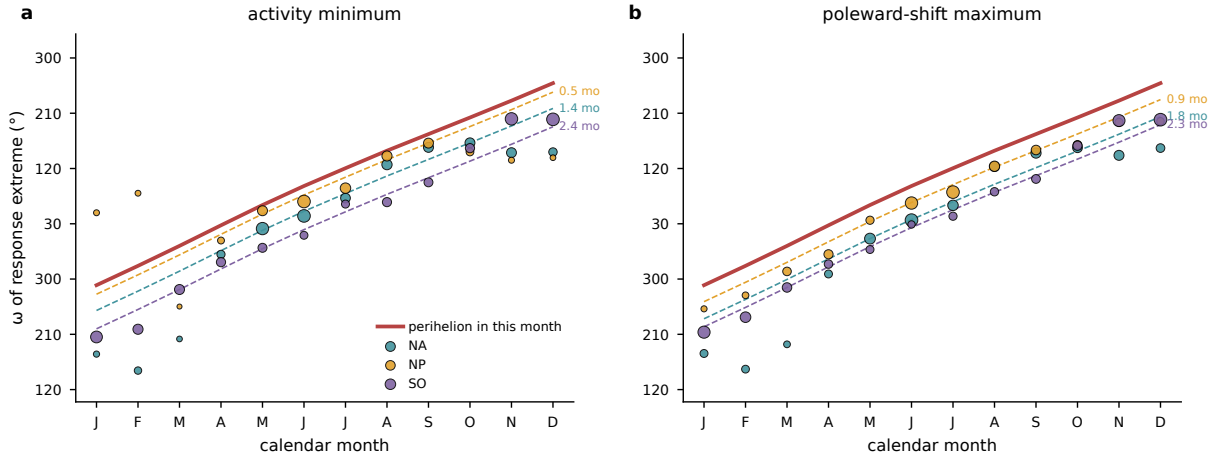


FIG. 8. Phase locking of the storm response to the perihelion calendar. For each calendar month and basin, the phase of the first harmonic across the 12 precession phases of (a) basin-mean storm days, shown as the ω of the activity minimum, and (b) the storm-day-weighted track latitude, shown as the ω of the poleward maximum. Marker size scales with the harmonic amplitude. The red line gives the ω that places perihelion in that calendar month, and dashed lines repeat it delayed by each basin's median lag over the amplitude-strong months (annotated in months).

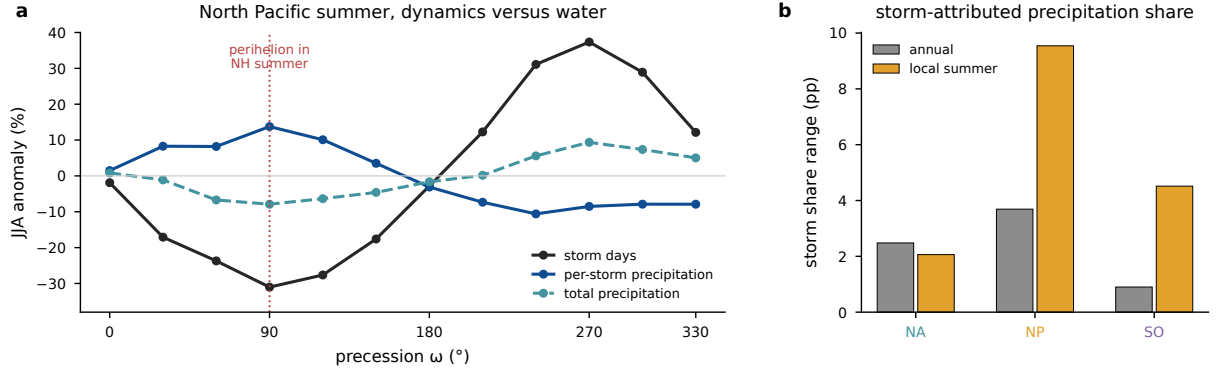


FIG. 9. Storm-attributed precipitation across the precession cycle. (a) North Pacific JJA percent anomalies from the phase mean of basin-mean storm days, per-storm precipitation, and total precipitation; the dotted line marks the phase with perihelion in NH summer. (b) Across-phase range of the storm-attributed share of precipitation, accumulated precipitation inside storm footprints divided by total accumulated precipitation (percentage points), for the annual mean and local summer in each basin.

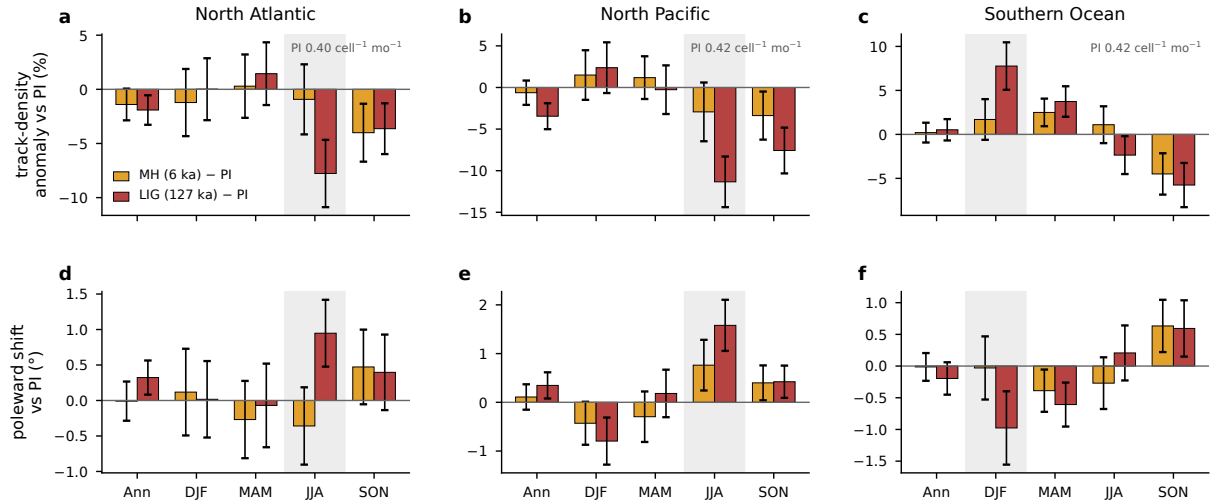
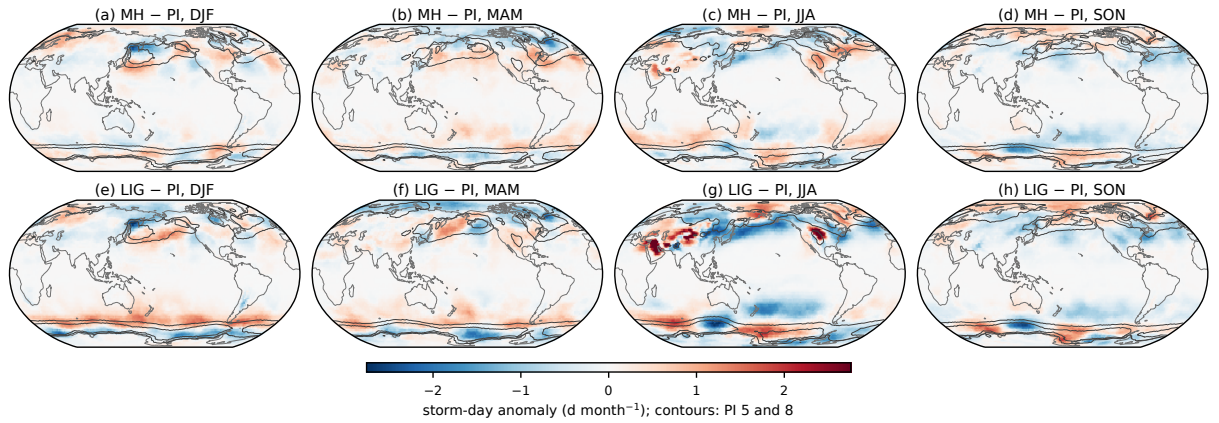


FIG. 10. Seasonal storm response of the real-orbit experiments, tracked with the same pipeline as the main ensemble and referenced to the preindustrial control. (a)–(c) Anomalies of basin-mean track density for the mid-Holocene (gold) and the Last Interglacial (red), as percentages of the preindustrial mean, for the annual mean and the four seasons, with error bars giving twice the standard error of the 30-year difference and the local summer shaded. Track density is normalized per T63 grid cell, and the annotated preindustrial values give the absolute scale in the units of Fig. 4. (d)–(f) The corresponding change in the track-point mean latitude, positive poleward.



517 FIG. 11. Storm-day anomalies of the real-orbit experiments relative to the preindustrial control, from the
 518 same storm-day product as the main ensemble. (a)–(d) Mid-Holocene minus preindustrial for DJF, MAM, JJA,
 519 and SON. (e)–(h) Last Interglacial minus preindustrial. Contours outline the preindustrial climatology at 5 and
 520 8 d month^{−1}. The isolated warm-season maxima over subtropical continental interiors reflect stationary surface
 521 heat lows rather than traveling cyclones.

527 TABLE 1. Three-basin comparison of the precessional storm response. Ranges are max minus min across
528 the twelve phases as a percentage of the phase mean. Storm and sensitive seasons refer to local winter and
529 local summer. The storm season is diagnosed from cyclogenesis counts, the sensitive season from the seasonal
530 storm-day range. The EGR response is the RMS p090–p270 difference over the basin sector relative to its
531 climatology, and the shear share is the projection of the vertical-shear factor onto the response pattern.

Basin	Storm season	Sensitive season	Storm-day range ann/summer (%)	Summer max/min ω (°)	EGR response summer (%)	Shear share (%)
North Atlantic	DJF	JJA	10 / 32	240 / 90	31	107
North Pacific	DJF	JJA	17 / 68	270 / 90	92	106
Southern Ocean	JJA	DJF	4 / 16	60 / 210	21	109

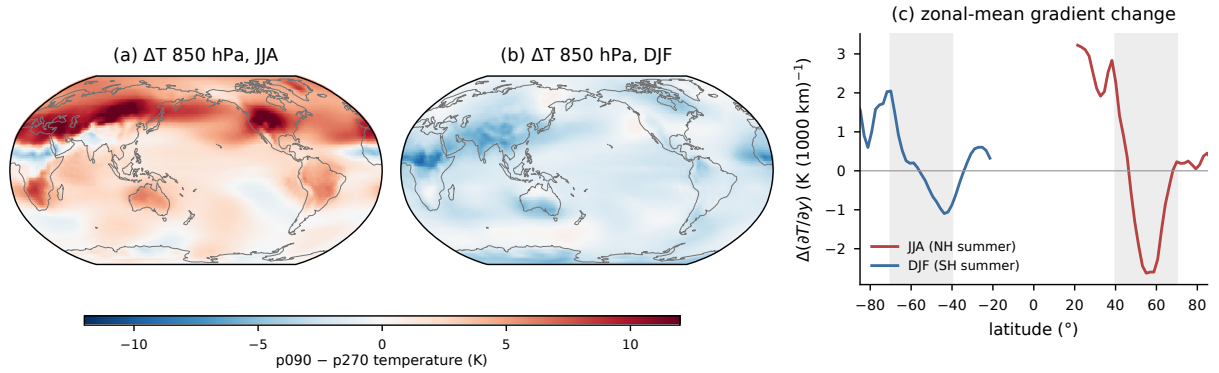


FIG. A1. Thermal footprint of the orbital end members. (a) p090 minus p270 temperature at 850 hPa for JJA and (b) for DJF (K). (c) Change in the zonal-mean meridional temperature gradient for JJA (red) and DJF (blue), with the 40–70° storm bands shaded. In the Northern Hemisphere negative values mark a steepening of the poleward temperature decline, and in the Southern Hemisphere positive values do.

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Data availability statement. The AWI-ESM simulations of the idealized precession ensemble are described in Shi et al. (2025). The cyclone-track database, the gridded storm statistics, and the analysis scripts underlying all figures will be archived and assigned a DOI upon acceptance. [TODO repository DOI.]

APPENDIX

Thermal footprint of the orbital end members

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